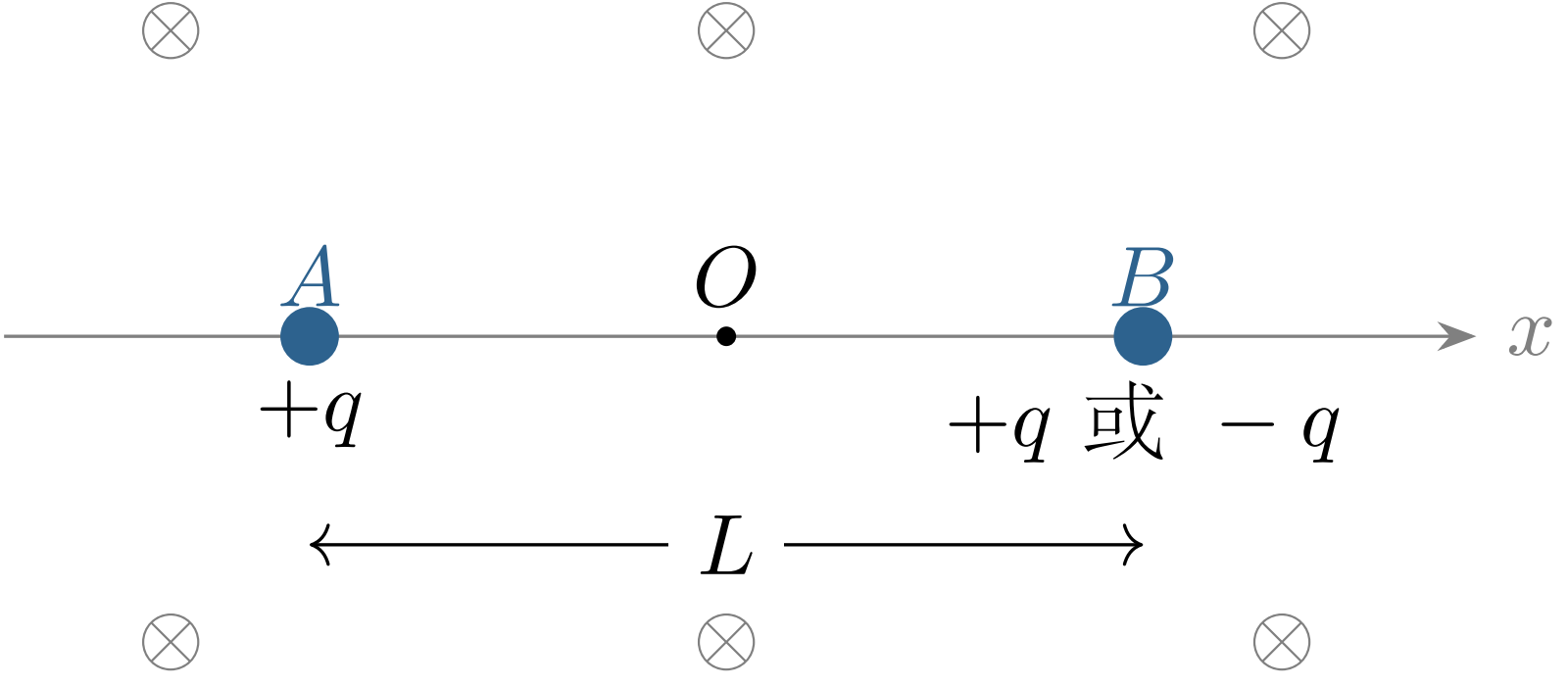
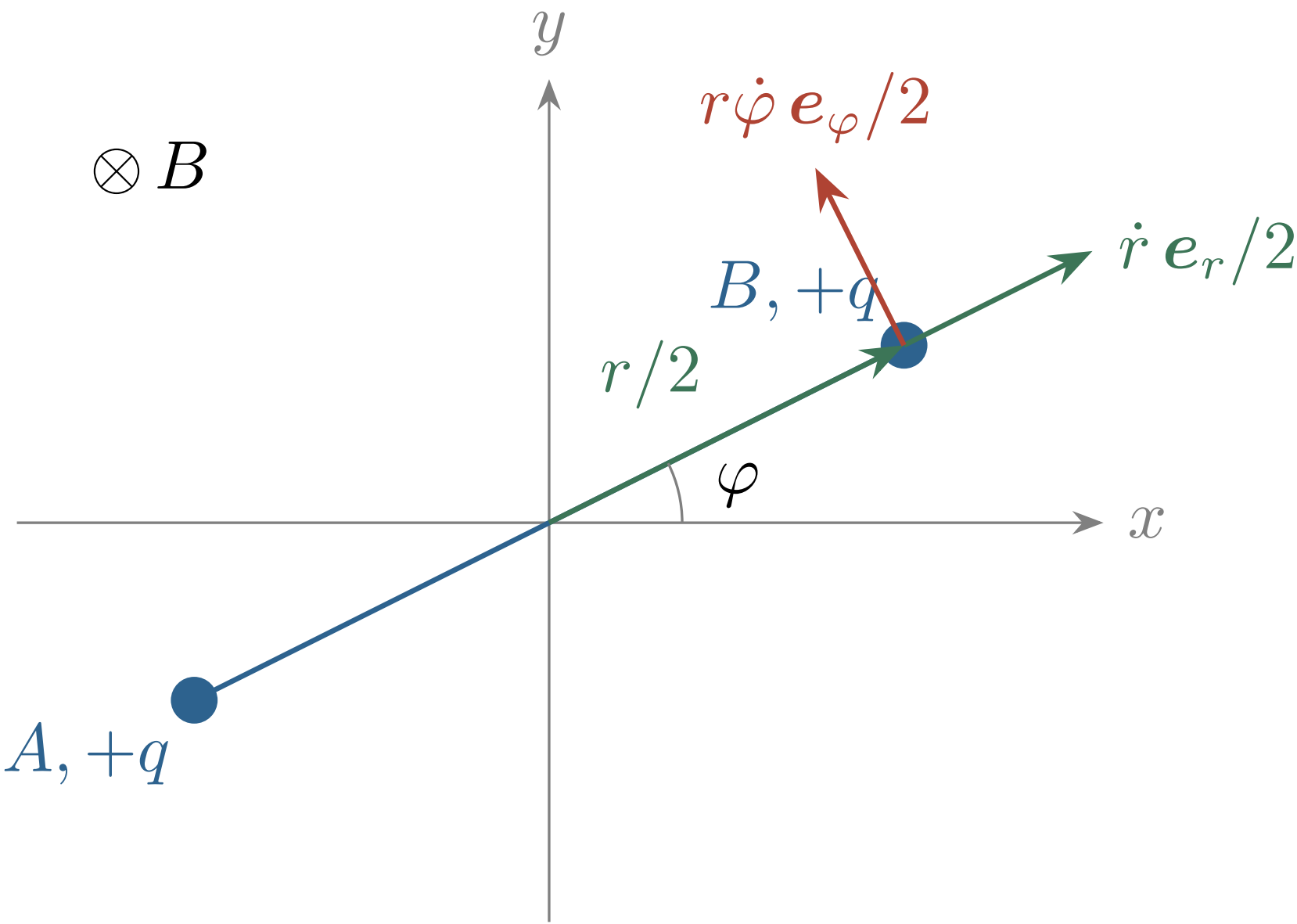
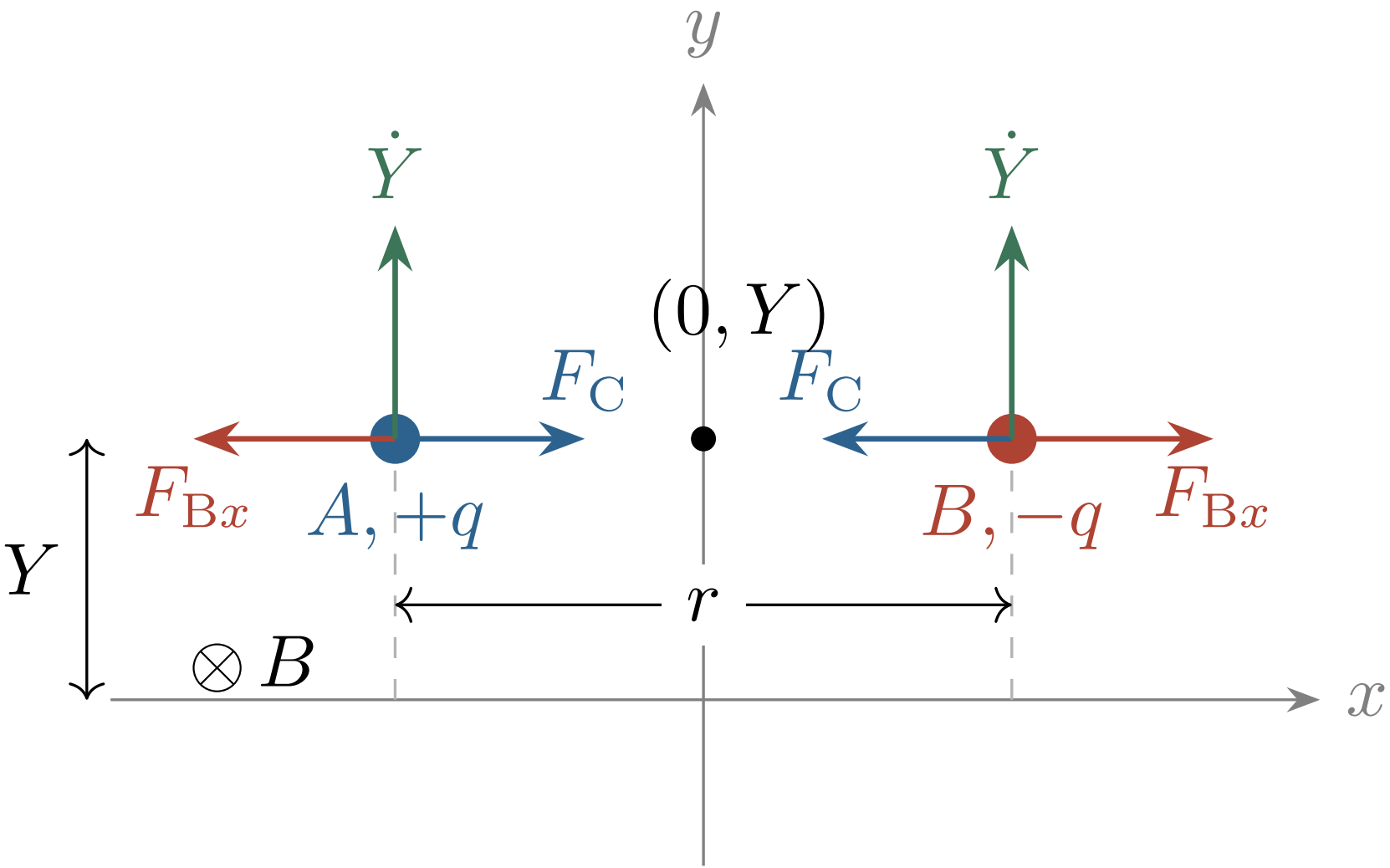
$$\boldsymbol{B} = -B\boldsymbol{e}_z, \quad \boldsymbol{v}_A(0) = \boldsymbol{v}_B(0) = 0$$



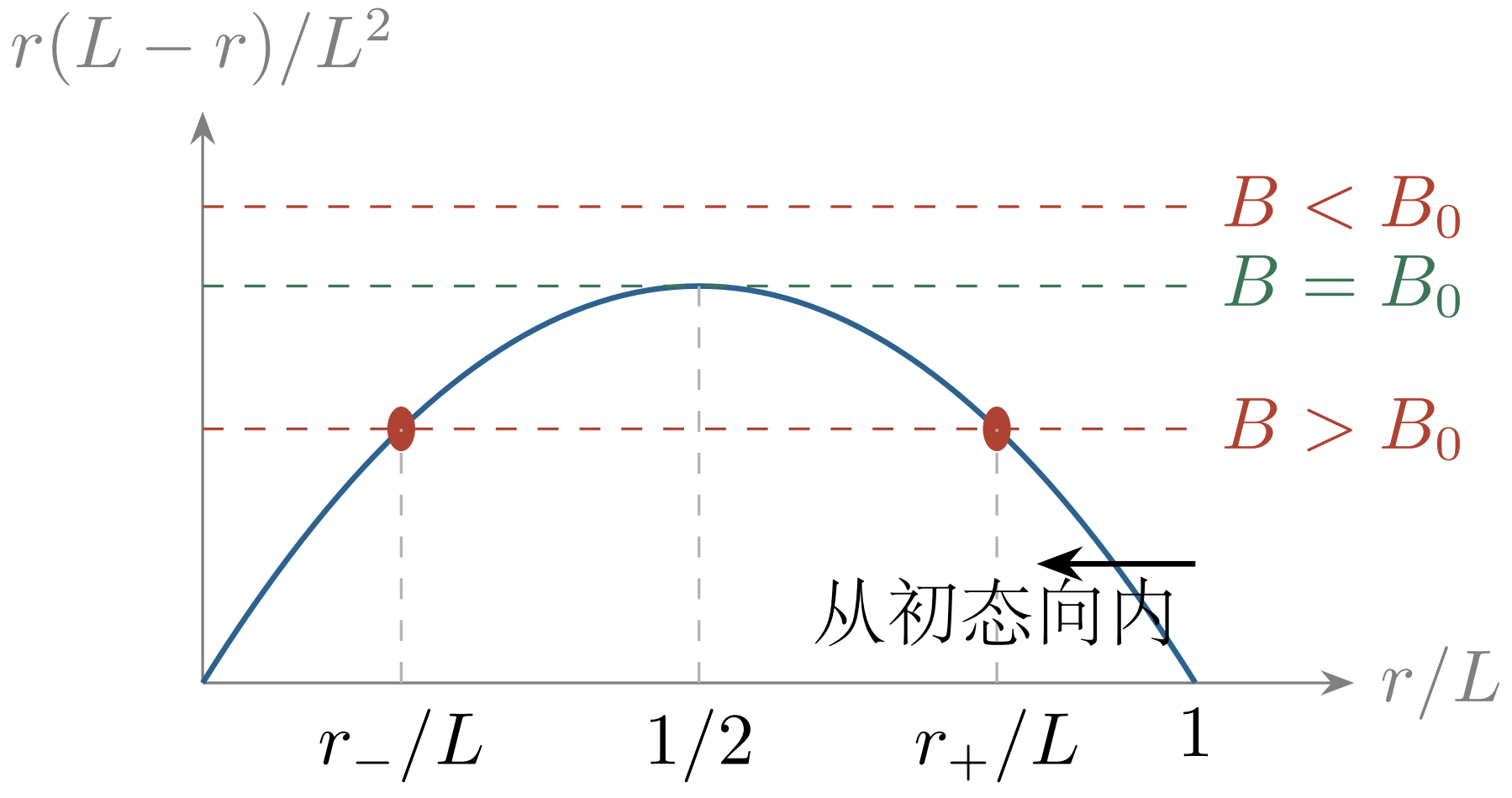


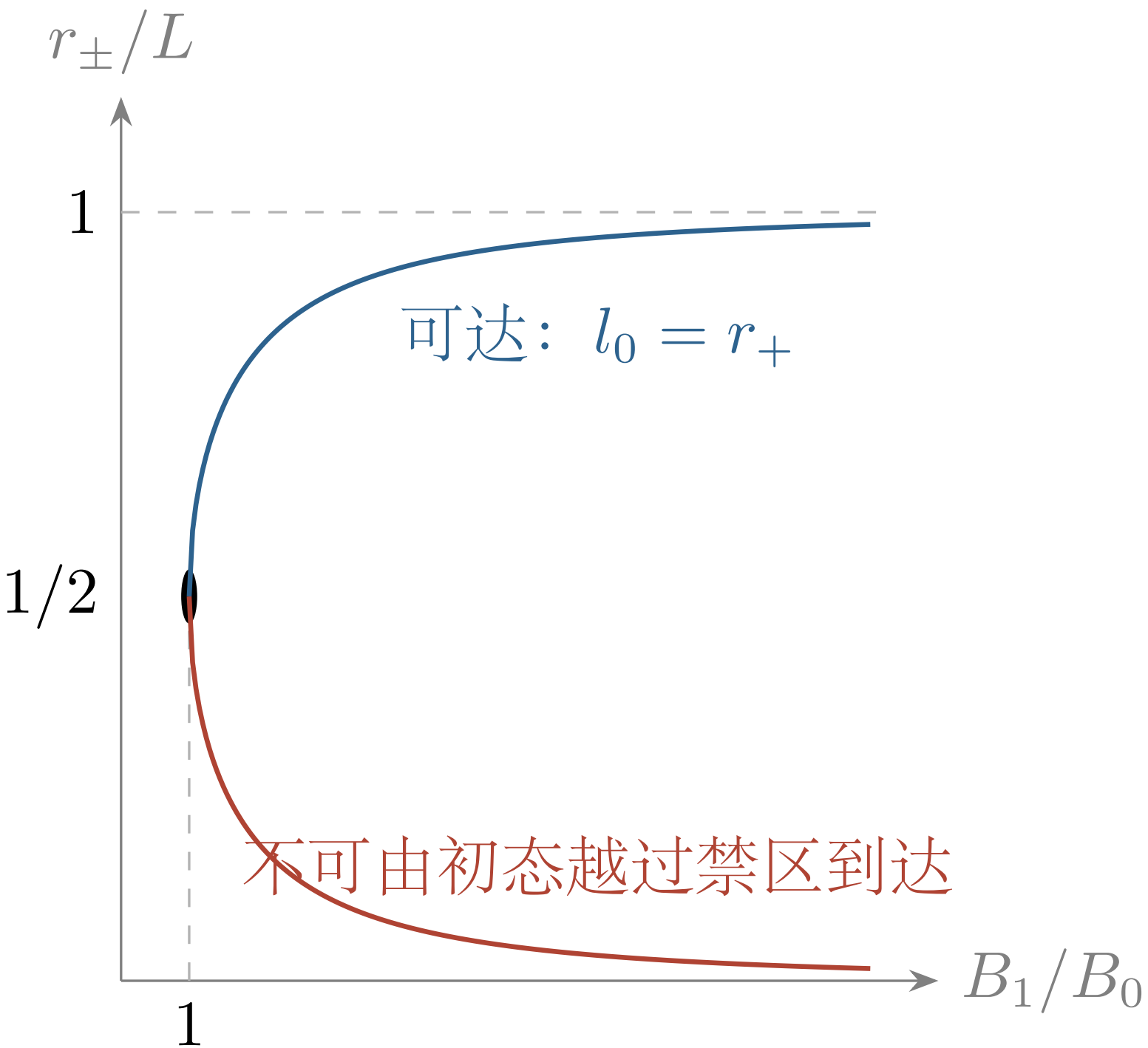
质心固定：$\boldsymbol{R} = 0$

最大间距处 $\dot{r} = 0$，但 $\dot{\varphi} \neq 0$



共同的 $+y$ 速度使磁力的 x 分量向外


$r(L-r)/L^2$
$B < B_0$
$B = B_0$
$B > B_0$
从初态向内
r/L
r_-/L
$1/2$
r_+/L
1


$r_\pm/L$
1
1/2
可达：$l_0 = r_+$
不可由初态越过禁区到达
1
B_1/B_0